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A SAM-Based Small-Scale Attention Network for Lunar Impact Crater Detection with DEM Data

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***Abstract*-The lunar polar regions, owing to their unique geographical location and the potential presence of water ice, have emerged as primary targets for recent lunar exploration missions by major spacefaring nations. Small impact craters, as the most ubiquitous and prominent surface features, offer invaluable scientific evidence for investigating both the internal and surface geological processes of planetary bodies. However, the inherent low solar elevation in these regions results in extensive permanently shadowed areas, which significantly hampers the acquisition of high-quality optical remote sensing imagery. Consequently, existing crater detection networks designed for optical images exhibit suboptimal performance when applied to digital elevation model (DEM) data. Moreover, the small scale and overlapping distribution of craters further constrain the effectiveness of current object detection methodologies. To address these challenges, this study proposes an object detection approach based on the Segment Anything Model (SAM). In the proposed framework, a SAM image encoder with frozen parameters is first employed to extract robust encoded features, which are then enhanced by a novel Small-Scale Attention (SSA) module designed to decode features specifically for small-scale targets. These refined features are subsequently integrated with a YOLO detection head to achieve high-precision crater detection. Experimental results demonstrate that the proposed method surpasses existing state-of-the-art object detection techniques in terms of detection accuracy. This research not only offers a new perspective for lunar polar region exploration but also broadens the applicability of SAM in remote sensing analysis.**

Keywords: Deep Learning, Segment Anything Model, DEM data, Lunar impact crater

I. INTRODUCTION

The lunar polar regions have garnered immense scientific interest due to their abundant mineral resources, potential reserves of water ice, unique geographical positioning, and the preservation of the solar system's impact history[1]. Consequently, these areas have become focal points for international lunar exploration and strategic high grounds for major spacefaring nations. Notably, the permanently shadowed regions (PSRs), situated at the bases of deep polar impact craters and characterized by persistently low solar elevation and obstructive crater walls, constitute critical targets for in-situ investigations. Such explorations are pivotal for capturing evidence of interactions among comets, asteroids, and the solar wind[2], and for elucidating the behavior of volatiles on the Moon and other airless bodies. Satellite impact observations have further revealed considerable deposits of water, ice and other volatiles in these areas[3], although questions regarding their state, composition, and abundance persist, necessitating further on-site analysis.

Impact craters serve as key geological markers that illuminate the evolutionary history of the Moon and its neighboring planetary systems[4, 5]. Their quantity, dimensions, and spatial distribution are critical parameters for determining the lunar surface's geological age and history[6]. Moreover, these craters provide essential ground reference points for spacecraft navigation and offer insights into Earth's impact history and the evolution of the solar system. Traditional crater identification methods, which relied heavily on manual annotation, are both labor-intensive and prone to subjective variability, often resulting in inconsistent outcomes.

In recent years, the advent of deep learning techniques has markedly advanced lunar crater detection[7, 8]. Conventional image processing and traditional machine learning methods have demonstrated limited performance when addressing the complexities of the lunar surface, whereas deep learning approaches have exhibited superior recognition capabilities[9]. Detection methods can be broadly categorized into those based on optical imagery and those utilizing digital elevation models (DEMs). Although optical-based methods offer high spatial resolution, they are highly susceptible to variations in illumination and the intricacies of lunar terrain. This limitation is especially pronounced in PSRs, where extremely low solar angles (1°–4°) and scarce secondary reflections impede effective optical remote sensing. While Bickel[10] has shown that deep learning algorithms can partially restore optical images of small craters, these techniques remain impractical for expansive PSRs. In contrast, DEMs constructed from high-density, high-quality altimetry data—such as those provided by the Lunar Orbiter Laser Altimeter (LOLA)—more accurately represent surface morphology and are therefore better suited for detecting small-scale impact craters in lunar PSRs[11, 12]. Recent studies have introduced innovative approaches such as LCDNet, a two-stage crater detection network that incorporates a retained negative sampling module and an adaptive non-maximum suppression module to mitigate issues related to incomplete annotations and overlapping craters[13]. Similarly, Juntao[14] et al. developed a terrain-aware network that leverages quantitative measures of terrain complexity to enhance model generalizability for small-scale crater detection. Additionally, the DeepMoon network—built upon a U-Net architecture and utilizing template matching combined with multi-scale feature fusion—has achieved high-precision crater detection,

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successfully identifying approximately 45% of previously unrecognized craters[15]. Despite these advancements, the aforementioned methods often exhibit limited generalizability, as they are typically confined to specific datasets.

More recently, the emergence of the Segment Anything Model (SAM) has heralded a new era in image segmentation. Trained on the expansive SA-1B dataset, SAM demonstrates exceptional generalization and zero-shot interpretation capabilities[16]. Nevertheless, its direct application to remote sensing imagery remains challenging due to the substantial differences between natural images and DEMs, resulting in only modest performance improvements on the latter.

To address these limitations, this study introduces a SAM-based approach for lunar crater detection. By leveraging SAM's robust feature extraction capabilities, the proposed method first acquires effective encoded features, which are subsequently refined through a bespoke Small-Scale Attention (SSA) module specifically designed to enhance the detection of diminutive craters. The experimental framework employs a dataset comprising 1,304 images of size 512×512 from the lunar polar PSRs, partitioned into 1,103 images for training and 201 images for testing. Comprehensive comparative evaluations underscore the superior performance of the proposed method over existing state-of-the-art detection techniques. The findings of this research not only offer a valuable reference for future lunar exploration endeavors but also extend the applicability of SAM in the domain of remote sensing analysis.

II. METHOD

A. Overview

Figure 1 illustrates the overall architecture, which is divided into two main components: the backbone and the head. In our approach, the image encoder from the Segment Anything Model (SAM) is employed as the backbone to extract deep encoded features, which are then processed by a detection head analogous to that of YOLOv11[17] to generate the final output. Notably, while the standard YOLOv11 detection head operates on features at scales of 1/4, 1/8, and 1/16 of the original resolution, we integrate a Small-Scale Attention (SSA) module and adjust the detection head to operate on feature maps at scales of 1/2, 1/4, and 1/8. This modification is specifically designed to enhance the network's capability in detecting small impact craters, thereby improving overall detection performance.

B. Backbone

Once the image is fed into the network, it is processed by the SAM image encoder with frozen parameters, yielding three hierarchical layers of encoded features. Given that SAM was pretrained on natural images, each layer is subsequently passed through an Adapter module to activate and adapt the features for the specific task, before being processed by the YOLOv11 C3k2[17] module.

C. Head and SSA module

For the deepest encoded features, convolutional up-sampling with skip connections is employed to progressively restore the spatial resolution, with a C3k2 module applied at each stage to preserve salient feature information. The resultant feature map, at one-quarter of the original resolution (denoted as F_2), is integrated with the original image and fed into the SSA module to obtain half-resolution features. As illustrated in Figure 2, the SSA module first applies a dual convolution followed by average pooling to extract features at half resolution, and then refines these features using channel and spatial attention mechanisms. The refined features subsequently serve as guiding skip connections to facilitate the up-sampling of F_2. After passing through an additional C3k2 module, the enhanced features are forwarded to the detector to produce the final output

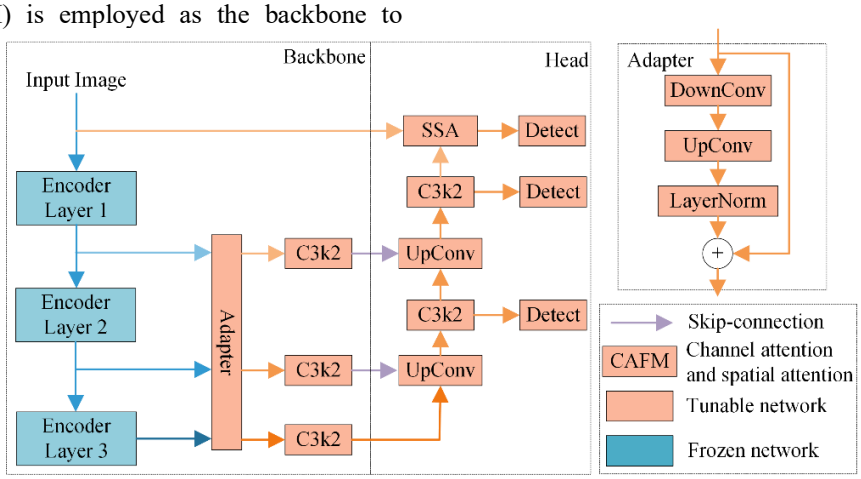


Figure 1. The pipeline of proposed SSA-SAM network.

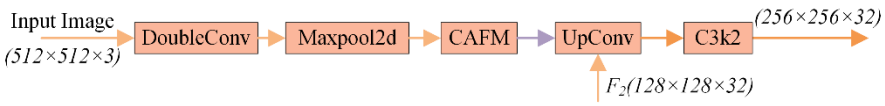


Figure 2. The pipeline of proposed SSA module.

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III. Experiment

A. Dataset

The experimental dataset used in this study is primarily derived from the original laser ranging dataset (EDR) and the calibrated geolocated dataset (RDR) released by the LOLA team. The RDR dataset integrates historical orbital data, attitude information, and lunar geolocation data from the Lunar Reconnaissance Orbiter (LRO). It is structured using the Planetary Data System (PDS) standard as fixed-length binary tables, accurately recording key parameters such as lunar surface elevation, laser pulse width, energy, and observation time. Data files are stored in fixed time intervals (approximately every two hours), ensuring high spatiotemporal continuity and reliability of the dataset.

In the data preprocessing pipeline, high-quality, high-precision DEMs were generated using the method proposed by Zheng[18, 19] et al. Based on these refined DEMs, we constructed a deep learning dataset specifically tailored for the detection of lunar impact craters in permanently shadowed regions. To enhance crater visibility for object detection tasks, the raw DEMs were transformed into pseudo-color images and subjected to low solar altitude illumination effects, making small craters more distinguishable. The dataset consists of 1,304 images, each with a resolution of 512×512 pixels, with impact craters annotated using the LabelMe tool. The dataset was further divided into a training set of 1,103 images and a test set of 201 images. The annotated results clearly delineate crater boundaries and morphologies, demonstrating the dataset's suitability and effectiveness for deep learning-based crater detection tasks. Figure 3 visually presents the dataset samples and labels.

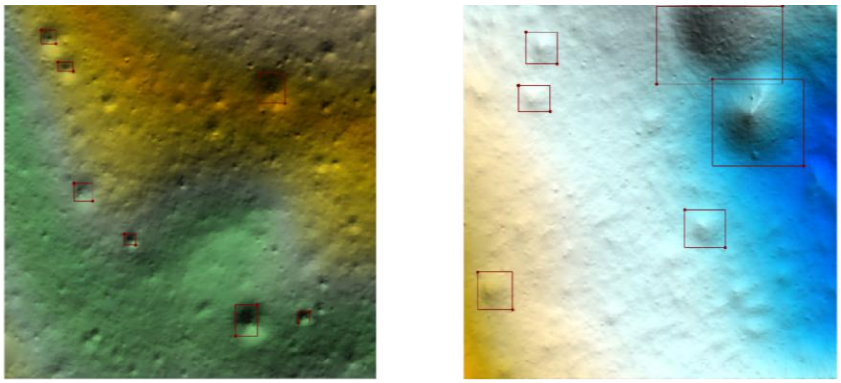


Figure 3. The samples of our dataset.

B. Experimental setup

In this study, the baseline network is defined as a network composed of the SAM image encoder and the YOLOv11 head. To validate the efficacy of the proposed method, experiments were conducted under the following configuration. The network was optimized using SGD with a momentum of 0.9, a weight decay of 0.00005, and an initial learning rate of 0.01. The architecture was implemented and trained within the PyTorch framework for 100 epochs. To bolster the model's generalization capabilities, the same data augmentation strategies employed in YOLOv11 were applied to the raw dataset. All experiments were executed on an Ubuntu 20.04 operating system, utilizing an NVIDIA A40 GPU equipped with 46GB of VRAM.

C. Experimental results

To validate the effectiveness of the proposed model, we conducted comparative experiments on various models and their enhanced versions, with results presented in Figure 4 and Table 1. The evaluation metrics include mAP_{50}, mAP_{95}, precision, and recall. A detailed comparative analysis of the experimental results is provided below.

As illustrated in Table 1, the performance of the YOLO series models (YOLOv5x, YOLOv8x, YOLOv10x, YOLOv11x)[20, 21] exhibits a gradual improvement. Specifically, YOLOv5x achieves mAP_{50} of 43.4, whereas YOLOv11x reaches an mAP_{50} of 48.9, with mAP_{95} increasing marginally from 17.0 to 17.6. However, when compared with the enhanced model proposed in this study, the performance gap is evident. Our model attains an mAP_{50} of 53.2—an improvement of 4.3 percentage points over the best-performing YOLOv11x—and delivers an outstanding mAP_{95} of 18.5 under a more rigorous IoU threshold, thereby significantly surpassing all versions of the YOLO series.

Table 1. Quantitative Comparison Table of Experimental Results.

	mAP_{50}	mAP_{95}	precision	recall
Yolo5x	43.4	17.0	44.6	43.8
Yolo8x	44.8	17.4	44.7	46.0
Yolo10x	45.1	16.4	45.4	44.8
Yolo11x	48.9	17.6	52.0	45.0
Baseline	47.6	16.3	51.1	45.3
Baseline+SSA	53.2	18.5	56.4	50.5

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Figure 4 visually presents the detection performance of various models on the test samples. It is evident that the YOLO series models yield relatively coarse detections and are prone to both missed detections and false positives, particularly when dealing with small targets and complex backgrounds. Although the Baseline model is capable of detecting some targets, the precision and completeness of its detection bounding boxes remain suboptimal. In contrast, the improved model proposed in this study demonstrates the ability to accurately detect targets even in more intricate terrain backgrounds, with results that closely align with the ground truth (GT). Compared with YOLOv11x, our model exhibits superior performance in terms of bounding box localization and target feature extraction.

Similarly, Figure 4 illustrates the detection outcomes of various models on the test samples, reiterating that the YOLO series models tend to produce coarse results with a high incidence of both omissions and false alarms, especially under conditions of small target sizes and complex backgrounds. While the Baseline model manages to detect certain targets, its bounding boxes still lack the required precision and integrity. In contrast, the improved model in this work achieves precise target detection in challenging terrain, yielding detection outputs that are more consistent with the GT. Relative to YOLOv11x, our model outperforms in both the localization of bounding boxes and the extraction of pertinent target features.

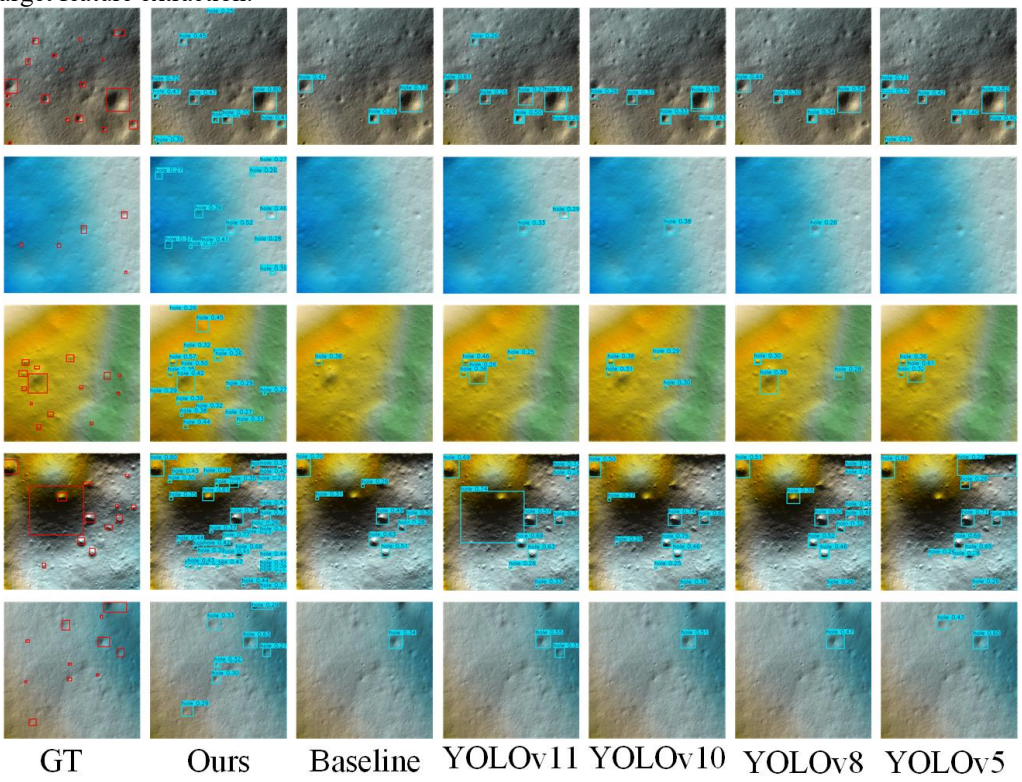


Figure 4. Qualitative Comparison of Experimental Results.

D. Ablation experiment

To further validate the specific contributions of each module to the model's performance, we conducted ablation experiments, with the results presented in Table 1. By comparing the accuracy between the Baseline and Baseline+SSA configurations, the effectiveness of the SSA module becomes evident. After incorporating the SSA module, mAP_{50} and mAP_{95} increased by 5.6% and 2.2%, respectively. Moreover, a comparison between the "Ours" and "Baseline" columns in Figure 4 demonstrates that, with the addition of the SSA module, the network achieves more accurate detection of small impact craters.

IV. Conclusion

This study addresses the task of detecting small impact craters in the permanently shadowed regions of the lunar poles by proposing an improved detection method based on high-quality DEM construction, which significantly enhances detection accuracy and robustness.

The main conclusions are as follows: First, to overcome the limitations of existing DEMs—characterized by low spatial resolution and insufficient capability for identifying small craters—we have constructed a high-quality DEM dataset with a 5-meter resolution. Compared to current 5-meter resolution products, this dataset exhibits superior accuracy in capturing terrain details and representing minor geomorphological features, thereby laying a crucial foundation for future research. Second, in light of the

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challenges faced by conventional methods, such as their strong reliance on optical imagery and limited robustness in detecting small craters, we propose a novel crater detection network, SSA-SAM. By integrating the SSA module, our approach demonstrates exceptional performance in detecting small and overlapping craters, significantly improving overall detection efficacy. Experimental results indicate that our method outperforms mainstream detection techniques in key metrics such as mAP_{50} and mAP_{95}.

Nonetheless, this study has some limitations; thus, future work will focus on expanding the scale of the high-quality DEM dataset and optimizing annotation quality to support more extensive crater detection tasks. Additionally, further in-depth geological analysis of the detected craters will be pursued to provide more precise data for understanding lunar geological evolution and resource exploration.

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